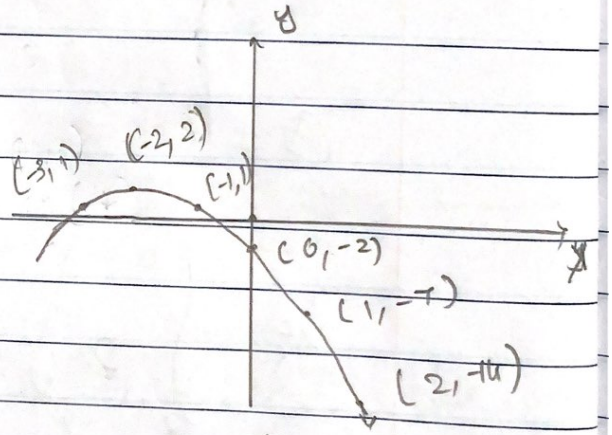
Exam 3

(1) $x = t - 2$, $y = 2 - t^2$

(a)

t	x	y
0	-2	2
1	-1	1
2	0	-2
3	1	-7
4	2	-14



(b) $t = 1$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2t}{1} = -2t$$

$t = 1$ Sub on equation $\Rightarrow$ $y =$

$$\boxed{m = -2}$$

$y - y_1 = m(x - x_1)$ where $(x_1, y_1) = (-1, 1)$

$\Rightarrow y - 1 = -2(x + 1)$

$\Rightarrow y = -2x - 2 + 1 \Rightarrow \boxed{y = -2x - 1}$

(2) $x = e^t - t$, $y = 4e^{t/2}$ $\quad 0 \le t \le 2$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$\frac{dx}{dt} = e^t - 1 \quad , \quad \frac{dy}{dt} = 4e^{t/2}\left(\frac{1}{2}\right)$$

$$\left(\frac{dx}{dt}\right)^2 = (e^t - 1)^2 \qquad \frac{dy}{dt} = 2e^{t/2}$$

$$\left(\frac{dy}{dt}\right)^2 = 4e^t$$

$$\left(\frac{dx}{dt}\right)^2 = e^{2t} - 2e^t + 1$$

$$L = \int_0^2 \sqrt{e^{2t} - 2e^t + 1 + 4e^t}\, dt$$

$$\Rightarrow L = \int_0^2 \sqrt{e^{2t} + 2e^t + 1}\, dt$$

$$\Rightarrow L = \int_0^2 (e^t + 1)\, dt$$

$$\Rightarrow L = \Big[e^t + t \Big]_0^2$$

$\Rightarrow \quad L = (e^2 + 2) - (e^0 + 0)$

$\Rightarrow \quad L = e^2 + 2 - 1$

$$\boxed{L = e^2 + 1}$$

(3) $y'' + y = \sin x$

(a)

$y = \sin x$

$y' = \cos x$

$y'' = -\sin x$

$\Rightarrow \therefore \quad y'' + y \neq \sin x$

(b) $y = \cos x$

$y' = -\sin x$

$y'' = -\cos x$

$y'' + y = \sin x$

$\therefore \quad -\cos x + \cos x \neq \sin x$

(c) $y = \frac{1}{2} x \sin x$

$y' = \frac{x}{2} \cos x + \sin x \frac{1}{2}$

$$y'' = -\frac{x}{2}\sin x + \cos x \frac{1}{2} + \frac{\cos x}{2}$$

$$\Rightarrow y'' + y = \sin x$$

$$\Rightarrow -\frac{x}{2}\sin x + \cos x + \frac{x}{2}\sin x \neq \sin x$$

(2) $y = -\frac{1}{2} x \cos x$

$$y' = +\frac{x}{2}\sin x - \cos x \frac{1}{2}$$

$$y'' = \frac{x}{2}\cos x + \frac{\sin x}{2} + \frac{\sin x}{2}$$

$$y'' + y = \frac{x}{2}\cos x + \sin x - \frac{1}{2} x \cos x$$

$$= \sin x$$

$\therefore y'' + y = \sin x$

when $y = -\frac{1}{2} x \cos x$

(4) $\frac{dy}{dt} = \frac{2t + \sec^2 t}{2y}$; $y(0) = -5$

$$2y\,dy = (2t + \sec^2 t)\,dt$$

$$\int 2y\,dy = \int (2t + \sec^2 t)\,dt$$

$$y^2 = t^2 + \tan t + C$$

$$y = \sqrt{t^2 + \tan t + C}$$

$y(0) = -5$

$$\Rightarrow -5 = \sqrt{0 + \tan(0) + C}$$

$$\Rightarrow -5 = \sqrt{C}$$

$$\Rightarrow \boxed{C = 25}$$

$$\boxed{y = \sqrt{t^2 + \tan t + 25}}$$

(5)

(a) $(3, -3)$

$$\tan\theta = \frac{-3}{3} \Rightarrow \theta = \tan^{-1}(-1)$$

$$\theta = 3\pi/4$$

$$x = r\cos\theta$$
$$y = r\sin\theta$$

$$\Rightarrow \quad x = r\cos(3\pi/4) \quad \Rightarrow \quad 3 = r\left(-\frac{\sqrt{2}}{2}\right)$$

$$\Rightarrow \boxed{\frac{-6}{\sqrt{2}} = r}$$

$$\therefore \boxed{(r, \theta) = (-6/\sqrt{2}, 3\pi/4)}$$

$$\boxed{(r, \theta) = (6/\sqrt{2}, 7\pi/4)}$$

(b) $(0, -5)$

$$\tan\theta = \frac{-5}{0} \Rightarrow \tan\theta = 0$$

$$\theta = 0 \text{ or } 2\pi$$

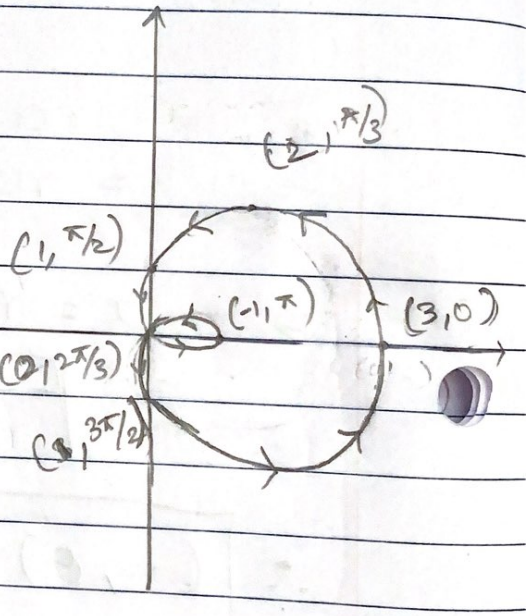
$$\Rightarrow \quad x = r\cos(0) \quad \Rightarrow \quad 0 = r(1) \quad \Rightarrow \boxed{r = 0}$$

$$\therefore \boxed{(r, \theta) = (0, 0)}$$

$(r, \theta) = (0, \pi)$

(6) $r = 1 + 2\cos\theta$

θ	$r = 1 + 2\cos\theta$
0	3
$\pi/3$	2
$\pi/2$	1
π	-1
$2\pi/3$	0
$3\pi/2$	1
2π	3



(7) $r = 1 + 2\cos\theta$, $\theta = \pi/6$

$r = 1 + 2\cos\theta$

$r' = -2\sin\theta$

$$\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2\sin\theta\sin\theta + (1+2\cos\theta)\cos\theta}{-2\sin\theta\cos\theta - (1+2\cos\theta)\sin\theta}$$

where $\theta = \pi/6$

$$\Rightarrow \frac{dy}{dx} = \frac{-2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + \left(1+2\left(\sqrt{3}/2\right)\right)\sqrt{3}/2}{-2\left(\frac{1}{2}\right)\left(\sqrt{3}/2\right) - \left(1+2\left(\sqrt{3}/2\right)\right)\left(\frac{1}{2}\right)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(-\frac{1}{2}\right) + \frac{\sqrt{3}}{2} + \frac{3}{2}}{\left(-\sqrt{3}/2\right) - \frac{1}{2} - \sqrt{3}/2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1+\sqrt{3}/2}{-\sqrt{3}-\frac{1}{2}} = \frac{\frac{2+\sqrt{3}}{2}}{\frac{-2\sqrt{3}-1}{2}}$$

$$\Rightarrow \boxed{\frac{dy}{dx} = \frac{2+\sqrt{3}}{-2\sqrt{3}-1}}$$

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - y_1 = \left(\frac{2+\sqrt{3}}{-2\sqrt{3}-1}\right)(x - x_1)$$

points :-

$x = r\cos\theta$, $y = r\sin\theta$

$\Rightarrow x = (1+2\cos\theta)\cos\theta$, $y = (1+2\cos\theta)\sin\theta$

$\Rightarrow x = (1+2\cos(\pi/6))\cos(\pi/6)$, $y = (1+2\cos(\pi/6))\sin(\pi/6)$

$\Rightarrow x = (1+\sqrt{3})(\sqrt{3}/2)$, $y = (1+\sqrt{3})(1/2)$

$\Rightarrow x = \sqrt{3}/2 + 3/2$, $y = 1/2 + \sqrt{3}/2$

$\therefore (x_1, y_1) = \left(\frac{\sqrt{3}+3}{2}, \frac{1+\sqrt{3}}{2}\right)$

$$\Rightarrow \boxed{y - \left(\frac{1+\sqrt{3}}{2}\right) = \left(\frac{2+\sqrt{3}}{-2\sqrt{3}-1}\right)\left(x - \frac{\sqrt{3}+3}{2}\right)}$$

(8) $r = 2$, $r = 4\cos 2\theta$

$\Rightarrow 4\cos 2\theta = 2$

$\Rightarrow \cos 2\theta = 1/2$

$\Rightarrow 2\theta = \pi/3 \Rightarrow \boxed{\theta = \pi/6}$

$$\text{Area} = \int_0^{\pi/6} \left(\frac{1}{2}\right)\left[(4\cos 2\theta)^2 - 4\right] d\theta$$

$$\text{Area} = \frac{1}{2}\int_0^{\pi/6} \left(16\cos^2(2\theta) - 4\right) d\theta$$

$$\text{Area} = 2\int_0^{16} \left(4\cos^2(2\theta) - 1\right) d\theta$$

(9) $$F = \int_a^b \rho g \,(\text{length of strip})\,(\text{depth})\, dy$$

$$F = \int_6^8 (1000)(9.8)(6)(20-y)\, dy$$

$$\boxed{F = 58800\int_0^8 (20-y)\, dy}$$

(11) $y = \sqrt{4x+6}$ on $[0,5]$

$$SA = \int_0^5 2\pi x \sqrt{1+[f'(x)]^2}\, dx$$

$$\frac{dy}{dx} = \left((4x+6)^{1/2}\right)'$$

$$\frac{dy}{dx} = \frac{1}{2}(4x+6)^{-1/2}(4)$$

$$\frac{dy}{dx} = \frac{2}{\sqrt{4x+6}} \Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{4}{4x+6}$$

$$SA = \int_0^5 2\pi x \sqrt{1 + \frac{4}{4x+6}}\, dx$$

$$SA = \int_0^5 2\pi x \sqrt{\frac{4x+6+4}{4x+6}}\, dx$$

$$SA = 2\pi \int_0^5 x \sqrt{\frac{4x+10}{4x+6}}\, dx$$

$$SA = 2\pi \int_0^5 x \sqrt{\frac{2x+5}{2x+3}}\, dx$$